

Worksheet 5B

Quiz 1: *What happens when we run?*

Program

```
(fun (foo n)
  (if (<= n 0)
    99
    (foo (sub1 n))
  )
)
(foo input)
```

Result

```
$ make foo.run
```

```
$ ./foo.run 100
```

```
$ ./foo.run 1000000
```

```
(fun (sum n acc)
  (if (= n 0)
    0
    (+ n (sum (sub1 n)))
  )
)
(sum input)
```

```
$ make sum.run
```

```
$ ./sum.run 100
```

```
$ ./sum.run 1000000
```

Quiz 2: *How many stack frames?*

How many stack frames are created when we run (sum 1000000)?

Quiz 3: *Modify assembly to reuse stack frame*

Given the source:

```
(fun (helper n acc)
  (if (= n 0)
    acc
    (helper (sub1 n) (+ acc n))))
```

Original Assembly

```
fun_start_helper:
push rbp                ; SETUP FRAME
mov rbp, rsp
sub rsp, 8*3

mov rax, [rbp - 8*-2]    ; IF (= n 0)
mov [rbp - 8*1], rax
mov rax, 0
cmp rax, [rbp - 8*1]
mov rcx, 3
mov rdx, 1
```

Optimized Assembly

```

cmove rax, rcx
cmovne rax, rdx
cmp rax, 1
je if_false_0 ; THEN BRANCH
if_true_0:
mov rax, [rbp - 8*-3]
jmp if_exit_0
if_false_0: ; ELSE BRANCH
mov rax, [rbp - 8*-2]
sub rax, 2
mov [rbp - 8*1], rax ; (sub1 n)
mov rax, [rbp - 8*-3]
mov [rbp - 8*2], rax
mov rax, [rbp - 8*-2]
add rax, [rbp - 8*2] ; (+ acc n)

push rax ; (helper ..)
mov rax, [rbp - 8*1]
push rax
call fun_start_helper
add rsp, 16
if_exit_0:

mov rsp, rbp ; TEARDOWN FRAME
pop rbp
ret

```

?

?

Quiz 4: Tail calls

A **tail call** is a function call that is the **last** thing executed in a function, i.e. there is **no more work** after the call returns.

Which of the two calls below are *tail calls*?

```

(fun (fac n acc)
  (if (= n 0)
    acc
    (if (= n 2)
      (* 2 (fac (sub1 n) (* acc n)))
      (fac (sub1 n) (* acc n))
    )
  )
)

```

Quiz 5: Which expressions can have tail calls?

- **Circle** the expression that *can* have tail calls
- **Cross out** the expressions that *cannot* have tail calls

```

e ::= n
    | true

```

```
| false
| input
| x
| (add1 e)
| (let (x e1) e2)
| (+ e1 e2)
| (= e1 e2)
| (if e1 e2 e3)
| (set x e)
| (block e1...en)
| (loop e)
| (break e)
| (print e)
| (call1 f e)
| (call2 f e1 e2)
```

Quiz 6: *Your turn!*

What is something you found confusing in today's lecture (or earlier)?